

High-bandwidth and High-efficiency MOS Mach-Zehnder Modulator based on InGaAsP Membrane/Si Waveguide with Capacitively-loaded Travelling-wave Electrodes

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Abstract We demonstrate InGaAsP/Si MOS Mach-Zehnder (MZ) modulators with capacitively-loaded travelling-wave electrodes. With a reduced RC constant, $V_{\pi}L$ -bandwidth trade-off is significantly improved, achieving bandwidth exceeding 60 GHz with $V_{\pi}L \sim 0.5$ Vcm, representing the highest reported bandwidth for MOS MZ modulators on Si photonics platforms. ©2026 The Authors

Introduction

In the last decade, data traffic in optical communication systems has increased rapidly. In particular, photonic integrated circuits (PICs) enabling high port density and low power consumption are required for optical transceivers in data-center networks. Among PIC platforms, Si photonics (SiPh) platforms have been widely used to satisfy these requirements, and various optical components have been integrated on SiPh platforms [1].

A Mach-Zehnder modulator (MZM) is one of the key photonic devices. Si-based MZMs have been extensively investigated over the past decade, because they can be fabricated using CMOS-compatible processes. In Si-based MZMs, the refractive index change Δn is induced by the free-carrier plasma dispersion effect, as follows;

$$\Delta n = -\frac{e^2 \lambda^2}{8\pi^2 c^2 \epsilon_0 n} \left(\frac{\Delta N}{m_e^*} + \frac{\Delta P}{m_h^*} \right)$$

where e is the elementary charge, λ is the wavelength of light, c is the speed of light, ϵ_0 is the dielectric constant, n is the refractive index, ΔN (ΔP) is the free electron (hole) density change, and m_e^* (m_h^*) is the electron (hole) effective mass. Among free-carrier modulation schemes in Si, carrier accumulation using a metal-oxide-semiconductor (MOS) capacitor is advantageous for achieving high-modulation efficiency. Si-based MZMs employing MOS structures have demonstrated high-modulation efficiency below a $V_{\pi}L$ (the product of the half-wave voltage V_{π} and the effective modulator length L) of 0.2 Vcm [2]. This value is several times lower than that of depletion-type Si-based MZMs using p-n junctions [3].

To further reduce $V_{\pi}L$, III-V/Si hybrid MOS MZMs have been proposed [4,5]. In these modulators, the free-carrier plasma dispersion effect in both the p-type Si layer and the n-type III-V layer

contributes to the refractive index change. Since the free-carrier plasma dispersion effect is inversely proportional to the effective mass, it is significantly stronger in n-type InGaAsP than in Si. In addition, the band-filling effect of InP-based direct bandgap materials also contributes to the refractive index change. As a result, the electron-induced refractive index change in InGaAsP is approximately 10 times larger than that in Si due to its small electron effective mass, enabling a $V_{\pi}L$ of 0.047 Vcm with an equivalent gate oxide thickness (EOT) of 5 nm [5].

III-V/Si hybrid MOS MZMs are capable of achieving both high-modulation efficiency and high-speed operation. However, the EO bandwidth of previously reported devices has been

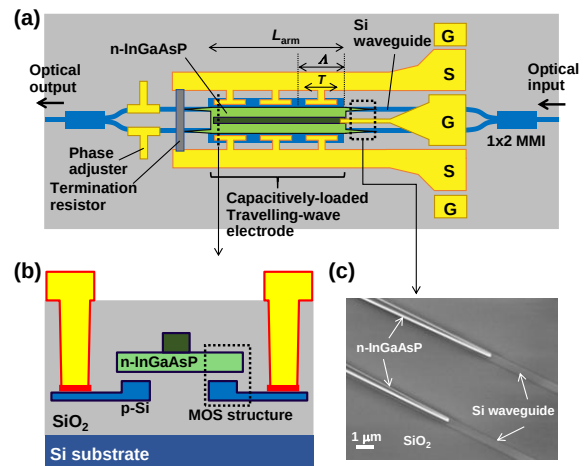


Fig. 1: (a) Schematic image of MOS MZ modulator with capacitively-loaded travelling wave electrodes. Here, L_{arm} , Δ , and T mean a total electrode length, a segment length, and a T-shaped electrode length, respectively. (b) Cross-sectional image of modulator section along dashed line in Fig. 1(a).

(c) The SEM image of optical coupling structures within dashed line in Fig. 1(a). Here, Si waveguide is covered with a thin SiO₂ layer.

limited to around 30 GHz, with a $V_\pi L$ of approximately 1.3 Vcm [6]. This limitation is attributed to large parasitic capacitance and resistance, as well as the use of lumped electrode structures that are not optimized for high-speed operation [4-6].

In this work, to address these limitations, we introduce a structure in which an InGaAsP membrane is shared between the two arms of the MZI. This configuration enables push-pull operation while reducing parasitic capacitance and resistance. By combining this structure with capacitively-loaded travelling-wave electrodes with a proper EOT, we successfully mitigate the trade-off between modulation efficiency and bandwidth, achieving an EO 3-dB bandwidth exceeding 60 GHz while maintaining a $V_\pi L$ as low as 0.5 Vcm. To the best of our knowledge, this is the highest EO bandwidth reported for MOS MZMs.

Experiments

Figure 1(a) depicts a plan-view schematic of an InGaAsP membrane/Si waveguide MOS MZM. An n-InGaAsP membrane is bonded on the two Si MZI arms. Capacitively-loaded travelling-wave electrodes (CLTWE) are employed as transmission lines to adjust impedance and achieve velocity matching between optical and electrical waves. In the CLTWE configuration, T-shaped electrodes are periodically arranged and optimized by tuning of filling factor (FF), which is defined as the ratio of the T-shaped electrode length (T) to the segment length (Λ). We also fabricated a primitive travelling-wave electrode (TWE) device as a reference to compare with reported lumped devices. Thermo-optic phase shifters are integrated for phase adjustment. Figure 1(b) shows a cross-sectional schematic of the MOS modulator section. The n-InGaAsP membrane overlaps the p-Si waveguide arms to form the MOS structure. By removing unnecessary slab regions in the Si waveguide, the structure minimizes parasitic capacitance that does not contribute to modulation. In addition, a single InGaAsP membrane is shared between the two arms, enabling push-pull operation while reducing parasitic resistance. Each T-shaped electrode of the CLTWE is formed on the p-Si waveguide via heavily p-doped Si regions to reduce contact resistance. Electrical isolation regions are also periodically formed in p-Si waveguide arms for the CLTWE.

For the fabrication of III-V/Si MOS MZM, a Si-on-insulator (SOI) wafer with SiPh circuits embedded by a SiO₂ cladding layer was prepared, which was planarized by chemical mechanical polishing. The thickness of the residual SiO₂ layer corresponds to EOT which determines a

capacitance of the MOS structure. An InP chip-to-SOI wafer bonding method was employed to form the III-V/Si MOS structure on the SiPh circuits. This bonding method has been developed for the fabrication of InP/Si heterogeneously integrated photonic devices [7,8]. Here, we chose an InGaAsP layer with a bandgap wavelength (λ_g) of 1.35 μm for enhancing the band-filling effect at a wavelength of 1.55 μm . The modulator region and taper structure for low-loss optical coupling between the Si waveguide and III-V/Si hybrid waveguide (Fig.1(c)) were sequentially defined using lithography and dry etching processes. The

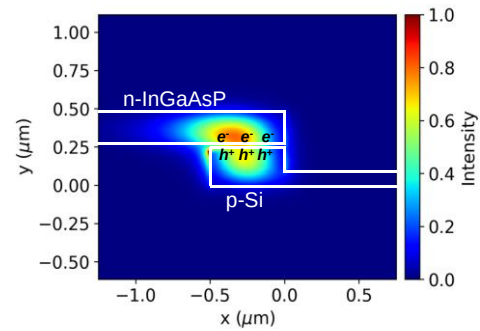


Fig. 2: Simulated cross-sectional optical mode-field pattern in the one side of MZ arms. Here, e^- and h^+ mean electron and hole, respectively.

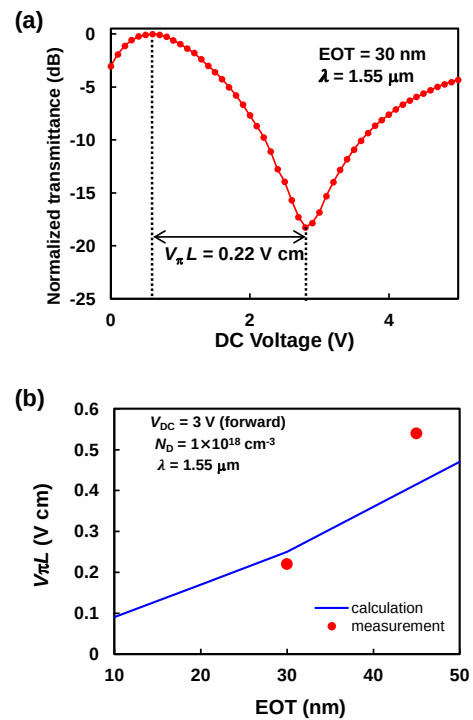


Fig. 3: (a) Normalized transmittance as a function of DC voltage V_{DC} applied to single-ended p-Si waveguide arm. (b) EOT dependence of modulation efficiency $V_\pi L$ using n-InGaAsP with a donor concentration N_D of $1 \times 10^{18} \text{ cm}^{-3}$. Here, V_{DC} is applied to p-Si with respect to n-InGaAsP.

optical coupling efficiency between the Si waveguide and III-V/Si hybrid waveguide is estimated to exceed 99% based on simulations. Finally, the modulator was completed by metal evaporation to form the CLTWE, RF termination resistors, and phase adjusters.

Results

Figure 2 shows the simulated cross-sectional optical mode-field profile in one arm of MOS MZM. In this configuration, the optical field is distributed across both the n-InGaAsP and p-Si layers as a fundamental mode. Under forward DC bias voltage V_{DC} , electrons and holes accumulate at the interface between n-InGaAsP and p-Si layers, referred to as the accumulation mode.

Figure 3 (a) shows the DC transmission property as a function of V_{DC} . Here, V_{DC} was applied to a single-ended p-Si waveguide of the MOS MZM. This device features a thin SiO_2 interlayer with an EOT of 30 nm, resulting in a low $V_{\pi}L$ of 0.22 Vcm. Figure 3 (b) shows the modulation efficiency $V_{\pi}L$ at a wavelength of $1.55 \mu\text{m}$ as a function of EOT under a V_{DC} of 3 V. Note that, in the CLTWE configuration, the $V_{\pi}L$ is calculated using an effective electrode length $FF \cdot L_{\text{arm}}$, while, in the TWE configuration, the effective length (L) equals to the L_{arm} itself. As a result, the $V_{\pi}L$ monotonically decreases with decreasing EOT, because the induced carrier sheet density is proportional to the capacitance of MOS structure. The experimental results exhibited good agreement with the calculations indicating high-modulation efficiency ($V_{\pi}L < 0.5 \text{ Vcm}$).

Figure 4 shows the frequency dependence of the electro-optic (EO) response under differential driving (Centre V_{DC} : 3 V) for the MOS MZM. The electrode lengths L_{arm} are 1 mm and 0.5 mm for CLTWE (FF: 60% and 80%) and TWE devices, respectively. In the CLTWE configuration, to achieve both high-bandwidth operation and low $V_{\pi}L$ ($\sim 0.5 \text{ Vcm}$), the EOT was optimized to 45 nm. Owing to employing travelling-wave electrodes, an EO 3-dB bandwidth of the TWE device (green line in Fig. 4) reached 35 GHz in accumulation-mode operation which is higher than that of lumped devices [4-6]. Furthermore, thanks to the use of CLTWE instead of TWE, the EO 3-dB bandwidth (blue and light blue lines in Fig. 4) was enhanced to 53 GHz for an FF of 80% (V_{π} : 6.75 V) and 61 GHz for an FF of 60% (V_{π} : 9 V) in accumulation-mode operation.

Figure 5 summarizes the relationship between $V_{\pi}L$ and EO 3-dB bandwidth previously reported MZMs on SiPh platforms. Toward the lower-right region, modulators with both lower $V_{\pi}L$ and higher bandwidth are located. Our results, indicated by red stars, are in the lower-right region,

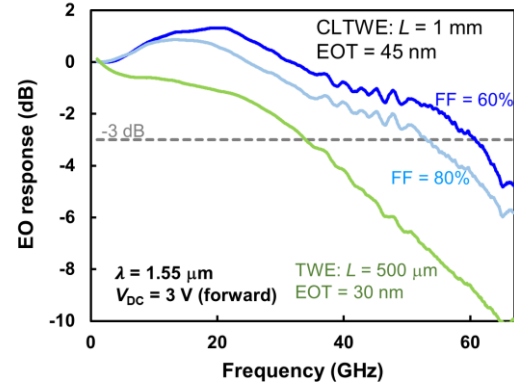


Fig. 4: Frequency dependence of EO response under accumulation mode and differential driving for devices with CLTWE (FF: 60% and 80%) and TWE.

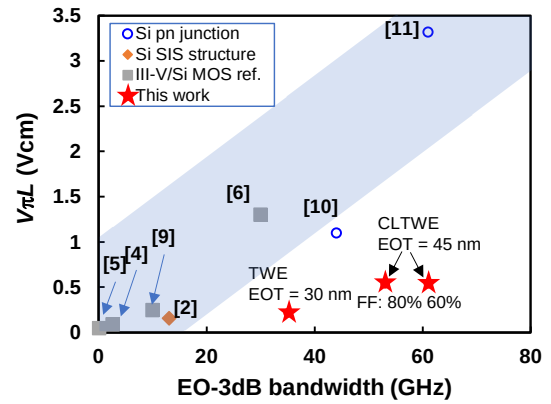


Fig. 5: Benchmarking in modulation efficiency and EO-3dB bandwidth for Si-based MZMs. The numbers in the figure indicate the references.

demonstrating both high bandwidth and low $V_{\pi}L$. In our devices, the reduction of the RC time constant and the introduction of capacitively-loaded travelling-wave electrodes significantly improve the inherent trade-off between $V_{\pi}L$ and bandwidth. Consequently, our devices successfully achieve a high EO 3-dB bandwidth exceeding 60 GHz while maintaining a low $V_{\pi}L$ of 0.5 Vcm. This bandwidth is the highest value in previously reported MOS MZ modulators on SiPh platforms, and it is sufficient for 100 GBaud operation.

Conclusions

We have demonstrated an InGaAsP/Si MOS MZM exhibiting a high EO 3-dB bandwidth exceeding 60 GHz while maintaining high-modulation efficiency ($V_{\pi}L \sim 0.5 \text{ Vcm}$). The reduction of the RC time constant, together with the introduction of capacitively-loaded travelling-wave electrodes, enables a significant improvement in the inherent trade-off between $V_{\pi}L$ and bandwidth. These results highlight the strong potential of III-V/Si MOS MZ modulators for realizing high-bandwidth, low-power optical modulators on SiPh platforms.

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